

Concentration Determination at a Countable Molecular Level in Nanofluidics by Solvent-Enhanced Photothermal Optical Diffraction

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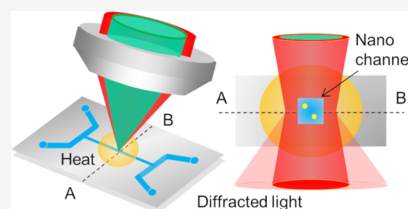


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ABSTRACT: Nanofluidic devices have become a powerful tool for extremely precise analyses at a single-molecule/nanoparticle level. However, a simple and sensitive molecular detection method is essential for nanofluidic devices because of ultrasmall volume (fL–aL). One such technology is photothermal spectroscopy (PTS), which utilizes light absorption and thermal relaxation by target molecules. Recently, we developed a photothermal optical diffraction (POD) detection method as PTS for nanofluidic devices. However, the detectable concentration range was in the order of μM (10^2 to 10^4 molecules), and further improvement in detection performance is strongly required. Here, we demonstrate solvent-enhanced POD with optimized experimental conditions and show its capability of concentration determination of nonfluorescent molecules in nanochannels at a countable molecular level. A relationship between the POD signal and thermal/optical properties of solvents is elucidated. We estimate the diffraction factor and photothermal factor of the solvent enhancement effect by thermal simulations and theoretical calculations. Experimental results show good agreement with the prediction, and the detection performance of the POD is successfully improved. At the optimized condition, we demonstrate the concentration determination with the limit of detection of 75 nM, which corresponds to an average of 10 molecules in a detection volume of 0.23 fL. Our sensitive nonfluorescent molecule detection method will be applied to a wide range of chemical/biological analyses utilizing nanofluidics.



The advent of microfluidic devices revolutionized analytical chemistry and pioneered high-throughput ultrasmall volume analyses.^{1,2} In recent years, further miniaturization to 10^1 to 10^3 nm (nanofluidic device) has gathered much attention because nanofluidic devices provide handling and chemical processing at a single-molecule/nanoparticle level.^{3,4} Several techniques utilizing nanostructures or nanochannels have been developed so far: sampling,⁵ preconcentration,⁶ trapping and sorting,⁷ chromatographic separation,^{8,9} and molecular sieving.¹⁰ By integrating these components, analyses of a countable number of proteins from a single cell have been realized.¹¹ Thus, nanofluidic devices are promising tools for a wide range of chemical/biological analyses.

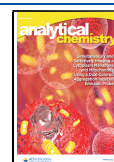
The number of target molecules in fL–aL samples is no more than 10^3 molecules even for a concentration range of μM to nM. Thus, ultrasensitive detection at a single or countable molecular level is essential for nanofluidic devices. Laser-induced fluorescence (LIF) has been widely used because of high sensitivity and selectivity.¹² Several papers report single-molecule detection and observation by LIF.^{13,14} However, few molecules have native fluorescence, and the labeling or tagging procedure is inevitable.¹⁵ To tackle this issue, various label-free detection methods have been developed: surface-plasmon resonance,¹⁶ micro-/nanoresonator,^{17,18} and nanopore/nanogap sensing.^{19,20} Although these methods have realized ultrasensitive detection at a single-molecular level, fabrication of complicated structures or molecular adsorption to a detection region is required.

Photothermal spectroscopy (PTS) provides label-free molecular detection without complicated structures or surface adsorption.²¹ PTS utilizes light absorption and thermal relaxation by target molecules (photothermal effect). The minute temperature change by heat is selectively measured by frequency-modulated synchronized detection, which enables sensitive molecular detection at a single or countable molecular level. For example, Tokeshi et al. realized sub-zeptomole Pb(II) octaethyl porphyrin detection in benzene solution by a thermal lens microscope (TLM).²² Gaiduk et al. demonstrated single-molecule imaging by a photothermal contrast microscope.²³ PTS also has been used for in vivo analyses and on-line detectors for chromatography.^{24,25} For micro-/nanofluidics, photothermal optical phase shift detection (POPS) was developed, and it realized protein molecule detection in a 2.1 μm deep channel at the limit of detection (LOD) of 30 molecules.²⁶ However, the detection performance of PTS largely degrades for nanochannels because of thermal diffusion from solutions to substrates (e.g., glass). For example, the difference of thermal property between the solvent and

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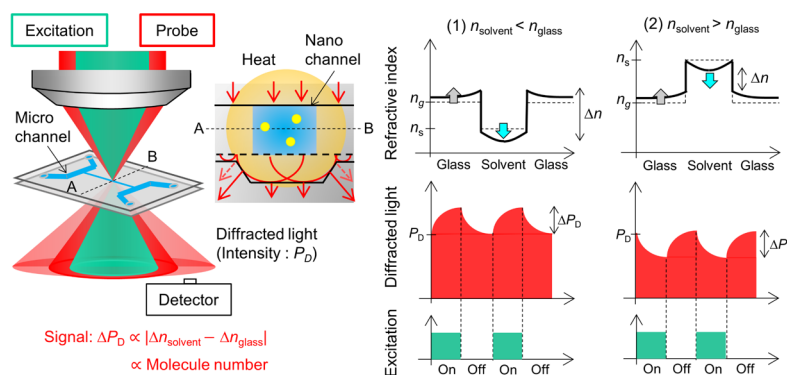


Figure 1. Detection principle of the POD. The photothermal effect of target molecules triggers temperature increase and refractive index change, leading to diffracted light intensity change (ΔP_D). For $n_{\text{solvent}} < n_{\text{glass}}$, Δn and P_D increase by excitation. For $n_{\text{solvent}} > n_{\text{glass}}$, Δn and P_D decrease by excitation.

substrate hinders the thermal lens formation or optical phase shift derived from the photothermal effect of target molecules. Thus, the detection principle of conventional PTS cannot be applied to ultra-small space. To overcome this physical limitation, we have developed the photothermal optical diffraction (POD) detection method, which utilizes the photothermal effect of target molecules and optical diffraction by a single nanochannel.²⁷ Different from other photothermal detection methods, the POD enables molecule/nanoparticle detection in 10^2 nm channels because thermal diffusion to substrates also contributes to the signal. For example, we demonstrated the detection and characterization of individual Au nanoparticles (20 nm in diameter) flowing in nanochannels.²⁸ We also realized chromatographic separation and concentration determination of nonfluorescent molecules in 10^2 nm channels.²⁹ The advantage of concentration determination in nanochannels over microchannels is accurate detection volume regulation from the diameter of an excitation beam spot and the cross-section area of a nanochannel, which enables accurate estimation of the number of molecules contributing the signal at a countable molecular level. However, the detectable concentration range of nonfluorescent molecules was in the order of μM (10^2 to 10^4 molecules) because of a low absorption coefficient compared with nanoparticles, and further improvement of detection performance is strongly required for single or countable molecule detection. Generally, the sensitivity of photothermal detection methods can be improved by utilizing organic solvents,³⁰ while the relationship between the thermal/optical properties of solvents and the sensitivity of the POD is unknown. Different from other photothermal detection methods, refractive indices of solvents also affect the POD signal, further complicating the effect of solvents. In addition to this, several unique liquid properties in nanochannels have been reported,^{31,32} which might affect the detection performance of the POD.

In this study, we report a solvent-enhanced POD detection method and concentration determination in 10^2 nm channels at a countable molecular level. First, we confirmed the detection principle of the POD and optimized experimental conditions such as diffracted light intensity and modulation frequency. Then, we investigated the effect of solvents on the detection performance of the POD. We estimated the diffraction factor and photothermal factor of the solvent enhancement effect by thermal simulations and optical calculations and compared calculated values with experimental

results. Finally, we evaluated the detection performance of the POD at the optimized condition.

EXPERIMENTAL SECTION

Principle and Optical Setup. The detection principle of the POD is briefly shown in Figure 1. Both probe beam and intensity-modulated excitation beam are focused on a nanochannel through an objective lens. The probe beam is diffracted by the nanochannel because of the difference of the refractive index between the solvent (n_s) and glass substrate (n_g). The diffracted light intensity (P_D) depends on the refractive index difference (Δn). Target molecules in the nanochannel absorb the intensity-modulated excitation beam, followed by thermal relaxation and periodical temperature changes of both the solvent and glass substrate (photothermal effect). Generally, the temperature coefficient of the refractive index (dn/dT) of solvents is negative, and dn/dT of glass substrates is positive; temperature change leads to either Δn increase or decrease depending on the refractive index of the solvent. The periodical change of Δn leads to the periodical diffracted light intensity change (ΔP_D), which is selectively measured by a photodiode and a lock-in amplifier. Thus, the concentration of the solution in the nanochannel is determined from ΔP_D . As an excitation beam, a 532 nm solid-state laser (Verdi-SM, Coherent Inc.) was used with 50 mW intensity under an objective lens (magnification: 20 $\times$, NA: 0.45). The intensity was modulated by a light chopper (5584A, NF Corporation, Japan). As a probe beam, a 633 nm He–Ne laser (25-LHP-925, Melles Griot Inc.) was used. A brief experimental setup is shown in the Supporting Information (Figure S1). Details of the detection principle and the optical setup is shown elsewhere.²⁷

Device Fabrication and Fluidic Control. We fabricated nanochannels on a fused-silica glass substrate (70 mm $\times$ 30 mm, VIOSIL-SQ, Shin-Etsu Chemical Co., Ltd., Japan) by electron-beam lithography and reactive-ion etching. The detailed fabrication process is shown in our previous report.³³ The fabricated nanochannel had a width and depth of 400 ± 20 nm and a length of 400 μm . For sample injection to nanochannels, we fabricated microchannels and inlet holes on another substrate by photolithography, reactive-ion etching, and drill. The fabricated microchannel had a width of 500 μm and a depth of 3 μm . These substrates were thermally bonded together at 110 $^\circ\text{C}$ for 3 h. Sample solutions were introduced to nanochannels via microchannels by a pressure-driven fluidic control system. Sample solutions in vials were pushed into

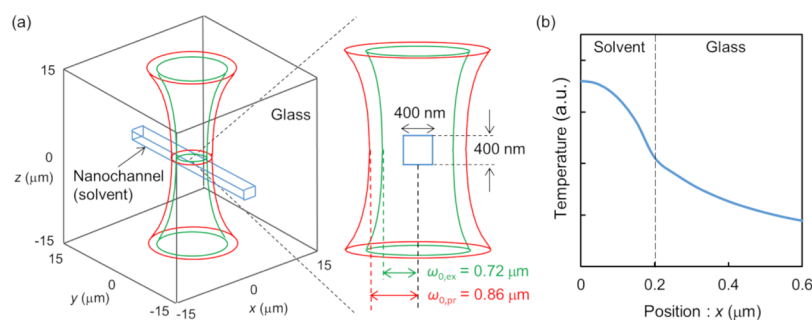


Figure 2. (a) Geometry of the nanochannel and lasers for the finite element analysis. (b) Example of calculated temperature distribution perpendicular to the nanochannel at $z = 0$ for aqueous solution.

microchannels through capillaries (0.26 mm inner diameter) by a pressure controller (MFCS-EZ, Fluigent Inc., France). For the replacement of sample solutions in nanochannels, the connected vial was exchanged with another vial, and the pressure was applied for a few minutes.

Sample Preparation. To investigate the solvent enhancement effect, we used six different solvents: water, ethanol, acetonitrile, hexane, toluene, and chloroform. Considering the solubility, Sunset Yellow FCF ($\epsilon = 4.7 \times 10^3 \text{ M}^{-1} \text{ cm}^{-1}$ for 532 nm excitation) was used as a solute for water and ethanol, and Sudan IV ($\epsilon = 1.8 \times 10^4 \text{ M}^{-1} \text{ cm}^{-1}$ for 532 nm excitation) was used as a solute for acetonitrile, hexane, toluene, and chloroform. To eliminate the effect of the different absorption coefficient, Sunset Yellow FCF in acetonitrile solution was prepared for normalization. All solvents and solutes were purchased from FUJIFILM Wako Pure Chemical Corporation, Japan. Before experiments, all the solutions were filtered through a $0.22 \mu\text{m}$ syringe filter to prevent nanochannel clogging. The concentration of each sample solution was checked by a UV–visible absorption spectrophotometer (Ultrospec 2100 pro, Biochrom Ltd., UK).

Estimation of the Solvent Enhancement Effect. Generally, the detection performance of PTS largely depends on the thermal/optical properties of the solvent such as a thermal diffusion coefficient. Furthermore, optical diffraction by nanochannels largely depends on the refractive index of the solvent in the nanochannel. Thus, we need to consider these two factors to realize a solvent-enhanced POD. For detection systems utilizing optical diffraction by a single nanochannel, the signal, the minute diffracted light intensity change (ΔP_D) is approximately expressed as follows³⁴

$$\Delta P_D \approx \frac{2P_D |\Delta n_s - \Delta n_g|}{|n_s - n_g|} = \frac{2P_D |\Delta n_s - \Delta n_g|}{|\Delta n|} \quad (1)$$

where n_s is the refractive index of the solvent in the channel, and n_g is the refractive index of the glass substrate. In this equation, Δn_s and Δn_g are determined by temperature change (ΔT) and the temperature coefficient of the refractive index (dn/dT), respectively. Thus, the detection performance of the POD will be improved by solvents with low Δn (diffraction factor) and high ΔT and dn/dT (photothermal factor). Although the diffraction factor can be estimated from eq 1, thermal simulation and calculation are necessary for the estimation of the photothermal factor.

Thermal Simulation. We performed a finite element analysis of thermal diffusion and calculated the photothermal factor of the solvent enhancement effect from temperature distribution. All simulations and calculations were performed

using COMSOL Multiphysics 5.4. The physical properties of the solvent and the glass substrate are according to the literature values,³⁰ and laser conditions were adjusted to those of actual experiments. Figure 2a shows the geometry of nanochannels and lasers for the calculation. The heat source distribution $Q(r, z, t)$ by the photothermal effect of target molecules in the nanochannel is described as follows³⁵

$$Q(r, z, t) = \frac{2Pa}{\pi^2 \omega_{ex}^2} \exp(-az) \exp\left(-\frac{2r^2}{\omega_{ex}^2}\right) \{1 + \cos(2\pi ft)\} \quad (2)$$

$$\omega_{ex}^2 = \omega_{0,ex}^2 \left(1 + \frac{z^2}{z_{r,ex}^2}\right) \quad (3)$$

where P [W] is the excitation laser intensity, α [m^{-1}] is the absorption coefficient of the sample, f [Hz] is the modulation frequency, $\omega_{0,ex}$ [m] is the waist radius of the excitation beam, and $z_{r,ex}$ [m] is the Rayleigh length of the excitation beam. Note that this thermal source distribution is valid only in the region of the nanochannel. The thermal distribution $T(r, z, t)$ was calculated by solving the heat conduction equation in the time-dependent model. Figure 2b shows the example of calculated temperature distribution at $y = 0$ and $z = 0$.

The POD signal is approximately proportional to the difference of average refractive index change between the solvent region and the glass substrate region. Therefore, the photothermal factor is calculated from the integrated value of the refractive index change weighted by probe laser intensity. The integral values for the solvent $S(t)_s$ and the glass $S(t)_g$ are described as follows

$$S(t)_{s/g} = \int \left(\frac{dn}{dT}\right)_{s/g} \{T(r, z, t) - T_0\} \left(\frac{\omega_{0,pr}}{\omega_{pr}}\right)^2 \exp\left(-\frac{2r^2}{\omega_{pr}^2}\right) dV \quad (4)$$

$$\omega_{pr}^2 = \omega_{0,pr}^2 \left(1 + \frac{z^2}{z_{r,pr}^2}\right) \quad (5)$$

where T_0 [K] is the initial value of temperature, $\omega_{0,pr}$ [m] is the waist radius of the probe beam, $z_{r,pr}$ [m] is the Rayleigh length of the probe beam, and dn/dT is the temperature coefficient of the refractive index. Note that the integral region is different for the solvent and the glass substrate. The relative photothermal factor $\Delta S(t)$ is defined as the difference of $S(t)_s$ and $S(t)_g$.

$$S(t) = |S(t)_s - S(t)_g|$$

$$= \int \left| \frac{dn}{dT} \right| \left\{ T(r, z, t) - T_0 \right\} \left(\frac{\omega_{0,pr}}{\omega_{pr}} \right)^2 \exp \left(-\frac{2r^2}{\omega_{pr}^2} \right) dV \quad (6)$$

In this equation, the value of dn/dT varies based on the integral region. Because the heat source distribution has a component that oscillates with the modulation frequency, $\Delta S(t)$ also oscillates at the same frequency. Therefore, we compared the maximum value of $\Delta S(t)$ for different solvents. We should emphasize that the above calculation after thermal simulation ignores temperature distribution and indirectly estimates the photothermal factor by integrated values. For the precise simulation of the absolute POD signal, we need to combine the thermal diffusion simulation and laser propagation simulation which will be included in future work, as in the present work, the main emphasis is on the estimation of the relative photothermal factor for different solvents with different thermal/optical properties.

■ RESULT AND DISCUSSION

Confirmation of the Detection Principle. In previous research, we verified the detection principle of the POD and demonstrated the concentration determination of nonfluorescent molecules in nanochannels.²⁷ However, it is unknown whether measured signals were derived from the change of the refractive index difference between the solvent and the glass substrate (Δn), which should be clarified for subsequent optimization. Therefore, we checked the signal generation mechanism by measuring signals of sample solutions with different refractive indices. Generally, dn/dT of solvents is negative, and dn/dT of the glass substrates is positive; refractive index changes of the solvent (Δn_s) and the glass substrate (Δn_g) are in opposite directions. As shown in Figure 1, when the refractive index of the solvent is lower than the glass substrate ($n_s < n_g$), Δn increases with temperature increase ($\Delta P_D > 0$). On the contrary, when the refractive index of the solvent is higher than the glass substrate ($n_s > n_g$), Δn decreases with temperature increase ($\Delta P_D < 0$). Thus, the relative phase of the signal for the excitation laser modulation will change 180° depending on whether the refractive index of the solvent is higher or lower than the refractive index of the glass substrate ($n = 1.46$). Figure 3 shows the time trace of signal intensity and phase when the solution in the nanochannel was replaced from hexane ($n = 1.38$) to toluene ($n = 1.50$). As a solute, Sudan IV was used for both solvents. The phase of the signal changed almost 180° when the solution in

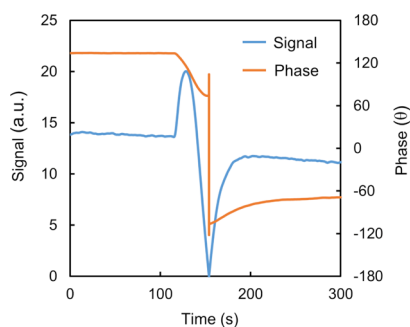


Figure 3. Time trace of signal intensity and phase when the solution in the nanochannel was replaced from hexane to toluene. Sudan IV was used as a solute.

the nanochannel was replaced. Also, the signal intensity became 0 at that time, suggesting that the sign of the ΔP_D was inverted. The signal increase at the initial state of the replacement might be due to the effect of mixing hexane and toluene. We also checked the time trace of signal intensity and phase when the solution in the channel was replaced from hexane ($n = 1.38$) to acetonitrile ($n = 1.34$), but no phase change was observed (Figure S2). From these results, we for the first time proved that the POD signal was generated from the change of Δn .

Optimization of Experimental Conditions. To improve the detection performance of the POD, we optimized the initial diffracted light intensity (P_D). The POD signal is the minute diffracted light intensity change (ΔP_D) by the photothermal effect of target molecules. A higher P_D leads to higher ΔP_D , while it also leads to larger background noise. We measured the signal-to-noise ratio (S/N) for different P_D , and the results are shown in Figure 4. We adjusted P_D by tuning

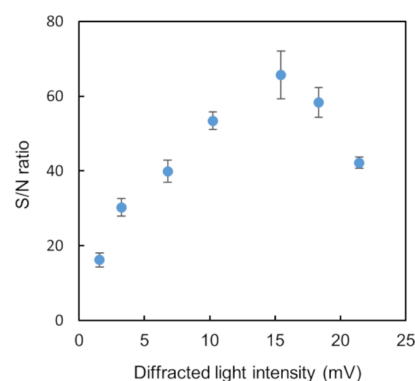


Figure 4. S/N ratio vs diffracted light intensity. Diffracted light intensity was measured by a lock-in amplifier at 1.0 kHz probe laser modulation.

the intensity of a probe beam laser, and the value was measured by a photodiode and a lock-in amplifier. As a test sample, Sunset Yellow FCF aqueous solution (100 μM) was introduced to nanochannels. The S/N ratio showed a maximum value around 15 mV. This result also shows that the noise derived from initial diffracted light (background noise) determines the detection performance of the POD. Note that P_D also depends on the size of the nanochannel and laser spot size; we need to adjust P_D measured by a photodiode and lock-in amplifier to the optimized value.

Furthermore, we optimized the modulation frequency of the excitation laser. Generally, higher modulation frequency leads to shorter laser irradiation time and signal intensity decrease. In contrast, lower modulation frequency leads to noise increase. Figure 5 shows the signal intensity and the S/N ratio versus modulation frequency. The signal intensity is inversely proportional to the modulation frequency, consistent with the general tendency of PTS.³⁰ The S/N ratio showed a maximum value around 1.1 kHz modulation. The thermal diffusion length also depends on the modulation frequency and can be expressed as follows

$$\mu = \sqrt{\frac{\kappa}{\pi C \rho f}} \quad (7)$$

where κ [W/m·K] is the thermal conductivity, C [J kg⁻¹ K⁻¹] is the specific heat, ρ [kg m⁻³] is the density of the solvent, and

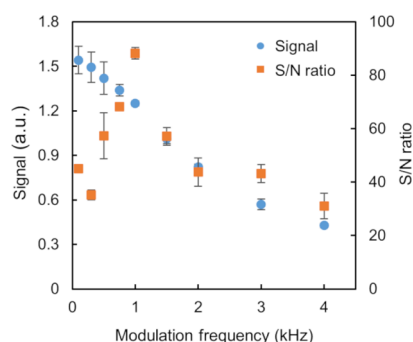


Figure 5. Signal intensity and S/N ratio vs modulation frequency. The diffracted light intensity was adjusted to 15 mV.

f [Hz] is the modulation frequency. The thermal diffusion length in water at 1.1 kHz modulation frequency is calculated to be 6 μm , much longer than the nanochannel width and depth. Thus, much of the generated heat diffuses to the glass substrate. From thermal simulation by COMSOL, the ratio of the total heat in water and glass substrate after thermal diffusion is estimated to be 1:150; the POD signal is generated by the temperature change of both solvent and glass substrate.

Investigation of the Solvent Enhancement Effect.

Next, we investigated the relationship between thermal/optical properties of solvents and the detection performance of the POD. As shown in eqs 1 and 6, the detection performance of the POD is determined by the diffraction factor ($1/\Delta n$) and the photothermal factor (ΔT and dn/dT). To confirm the signal enhancement by these factors, we evaluated the sensitivity of the POD for six different solvents: water, ethanol, acetonitrile, hexane, toluene, and chloroform. Table 1 shows

Table 1. Thermal/Optical Properties of the Selected Solvents²⁹

solvent	n_s	dn/dT [$\times 10^{-4}/\text{K}$]	κ [$\text{W}/\text{m}\cdot\text{K}$]	α [$\times 10^{-8} \text{ m}^2/\text{s}$]
water	1.33	−0.91	0.60	14.3
ethanol	1.36	−4.00	0.17	8.8
acetonitrile	1.34	−4.55	0.21	20.7
hexane	1.38	−5.20	0.12	8.4
toluene	1.50	−5.68	0.14	9.2
chloroform	1.44	−6.03	0.12	8.1

the thermal/optical properties of the selected solvents, where n_s is the refractive index, dn/dT [K^{-1}] is the temperature coefficient of the refractive index, κ [$\text{W}/\text{m}\cdot\text{K}$] is the thermal conductivity, and α [m^2/s] is the thermal diffusivity. The photothermal factor of the enhancement effect can be estimated from the sensitivity difference between water, ethanol, and acetonitrile solutions because these solvents have almost the same refractive indices but different dn/dT and thermal properties. The diffraction factor of the enhancement effect can be estimated from the sensitivity difference between hexane toluene and chloroform solutions because these solvents have similar dn/dT and thermal properties but different refractive indices. The initial diffracted light intensity (P_D) largely changes for solvents with different refractive indices. Therefore, P_D was adjusted to optimized value (15 mV) for all solvents by tuning probe laser intensity.

Figure 6 shows the calibration curves for four organic solvents. In repeated experiments, we observed the variation of blank signals (0 μM) from day to day and device to device, the

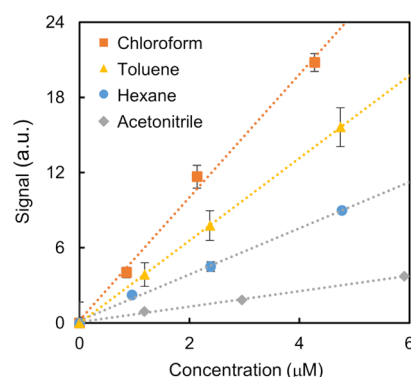


Figure 6. Calibration curves for four organic solvents. Sudan IV was used as a solute.

reason of which is under investigation. However, the variation showed little effect on sensitivity. Therefore, the blank signal was subtracted for each measurement. The signal intensity linearly increased with the concentration for each solvent. Figure 7 shows the normalized sensitivity (slope of the

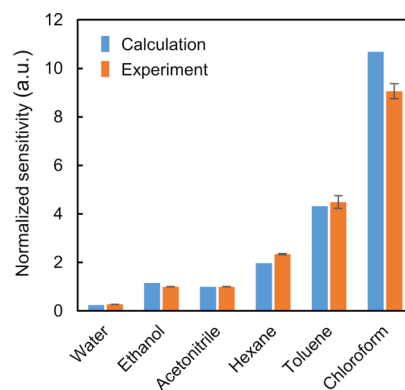


Figure 7. Comparison of sensitivity for each solvent. Each sensitivity was normalized based on the sensitivity of acetonitrile.

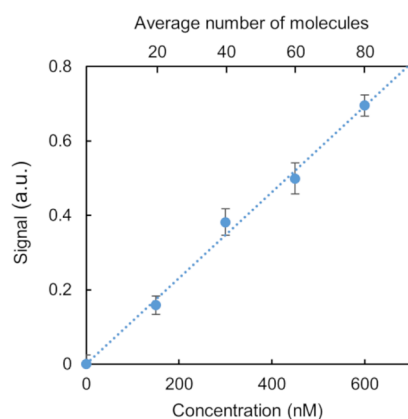
calibration curve) for each solvent. The calculated values were derived from literature values and thermal simulations.³⁰ Experimental results were almost consistent with the calculated values; unique liquid properties of water in nanochannels have little effect on the POD signal. Ethanol and acetonitrile solution showed almost the same enhancement, and the sensitivity was 3.6–3.7 times higher than the aqueous solution. These results suggest that the enhancement effect is derived from the photothermal factor (ΔT and dn/dT). Obtained enhancement effects seem relatively small, judging from high ΔT ($\times 1.7$ – 1.8) and dn/dT ($\times 4.4$ – 5.0) of ethanol and acetonitrile. As explained in the Experimental Section, the temperature change of glass substrates also contributes to the signal, which cannot be enhanced by solvents. Thus, the signal from glass substrates decreases the apparent solvent enhancement effect. Toluene and chloroform solutions showed 1.9 times and 3.8 times higher sensitivity than hexane solution, respectively, suggesting the enhancement effect by the diffraction factor ($1/\Delta n$). The enhancement factor of solvents relative to water is summarized in Table 2. Each diffraction factor was calculated from eq 1, and each photothermal factor was calculated from eq 6. Compared with the aqueous solution, more than 30 times sensitivity improvement was realized by organic solvents. Furthermore, obtained enhance-

Table 2. Summary of the Enhancement Factor of Five Different Solvents Relative to Water

solvent	diffraction factor	photothermal factor	total (calc.)	total (exp.)
ethanol	×1.3	×3.8	×4.8	×3.6
acetonitrile	×1.1	×3.8	×4.2	×3.7
hexane	×1.5	×5.4	×8.3	×8.6
toluene	×3.2	×5.7	×18.2	×16.5
chloroform	×7.5	×6.0	×45.0	×33.0

ment factors are different from those for conventional photothermal detection methods,³⁰ suggesting that both the diffraction factor and photothermal factor contribute to the sensitivity of the POD. The difference between calculations and experiments might be derived from the approximations in simulations and calculations.

Evaluation of Detection Performance. Finally, we evaluated the detection performance of the POD at the optimized condition. Although toluene and chloroform solutions showed high sensitivity among six different solvents, signal variation and noise also increased. This might be due to the high sensitivity of the diffracted light intensity for unexpected refractive index fluctuation. Therefore, we evaluated detection performance for hexane solutions. Figure 8 shows the calibration curve for the Sudan IV hexane solution

**Figure 8.** Calibration curve for Sudan IV hexane solution. The average number of molecules was calculated from the concentration and the detection volume.

in the nanochannel with a width and depth of 400 nm. The detection volume can be estimated to be 0.23 fL from the excitation beam spot diameter (1.4 μm) and the cross-section area of the nanochannel, and the average number of molecules in the detection volume is calculated from the concentration. The concentration determination was realized in the order of nM (10–100 molecules). The calibration curve showed good linearity with the concentration. The LOD is defined as a concentration that gives a signal equivalent to blank signal +3 σ , and σ is calculated from the signal fluctuation of lowest

concentration solutions. The calculated LOD was 75 nM, which corresponds to an average of 10 molecules in the detection volume of 0.23 fL. By the enhancement effect of organic solvents and the optimized experimental condition, we for the first time realized label-free concentration determination in a 10² nm channel at a countable molecular level. We also evaluated the detection performance for Sunset Yellow aqueous solutions. The calculated LOD was 2.0 μM which corresponds to an average of 260 molecules in the detection volume. The difference in the LODs between Sudan IV and Sunset Yellow is explained by the solvent enhancement effect and the difference in the molecular properties. Although hexane solutions give 8.6 times higher sensitivity than aqueous solutions, signal fluctuation (noise) also increases. Moreover, the difference in signal intensity between Sudan IV and Sunset Yellow was higher than expected from the absorption coefficient. The reason is under investigation. The detection performance compared with other photothermal detection methods for micro-/nanofluidics is presented in Table 3. The TLM achieved 1 order of magnitude lower LOD than that of the POD, and the UV-POPS realized ultrasensitive concentration determination without the solvent enhancement effect. However, these methods were applied to microchannels and cannot be applied to nanochannels because of sensitivity decrease by thermal diffusion. The POD realized concentration determination at a countable molecular level in nanochannels and can be applied to ultrasensitive volume analyses utilizing nanofluidics. Moreover, concentration determination measures the average number of molecules in the detection volume, not the total number of molecules passing through the detection region. In other photothermal detection methods for microfluidics, the detection volume is calculated from the confocal volume of the excitation laser, which has a possibility to underestimate the number of molecules contributing to the signal. In the POD, the detection volume is calculated from the excitation beam spot diameter and the cross-section area of nanochannel (=total light irradiation volume), which makes accurate estimation of the number of molecules contributing to the signal.

CONCLUSIONS

Our detection method realized label-free concentration determination in a 10² nm channel at a countable molecular level by optimizing experimental conditions and utilizing the solvent enhancement effect. We for the first time elucidated the relationship between thermal/optical properties of solvents and the detection performance of the POD, which will be useful for the application to normal-phase chromatography or solvent extraction. Although our detection method showed high detection performance, further sensitivity improvement is still necessary for single-molecule detection. As mentioned in the Optimization Section, the bottleneck for improving the detection performance is large background noise derived from the initial diffracted light. This problem might be solved by introducing interference optical systems which minimize the

Table 3. Comparison of Photothermal Detection Methods for Micro/Nanofluidics

	solvent	dn/dT [$\times 10^{-4}/\text{K}$]	analyte (absorbance [$\text{M}^{-1} \text{cm}^{-1}$])	channel size	LOD [molecules]
TLM ²²	benzene	−6.52	Pb(II) octaethyl porphyrin ($\epsilon = 3.2 \times 10^4$)	150 $\mu\text{m} \times 100 \mu\text{m}$	0.34
POPS ²⁶	water	−0.91	BSA ($\epsilon = 2.2 \times 10^4$)	100 $\mu\text{m} \times 2.1 \mu\text{m}$	30
POD	hexane	−5.20	Sudan IV ($\epsilon = 1.8 \times 10^4$)	400 nm $\times$ 400 nm	10

effect of background noise. Optimizing the material of the substrate is another way of sensitivity improvement. The sensitivity of the POD also largely depends on the thermal/optical properties of substrates. Utilizing materials with low thermal conductivity or high dn/dT such as TiO_2 or CaF_2 will greatly improve detection performance. Thus, our detection method has the potential for further improvement and will be widely used for several analyses utilizing nanofluidic devices.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.analchem.0c02024>.

Experimental setup and supporting data for confirmation of the detection principle (PDF)

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Notes

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